

(46 marks)

Write the answer to each question on *the question paper*. Show all working.

Question 1**(11 marks)**

- a) Find the value of $13^{-1.3}$ correct to three decimal places.

(1)

- b) Convert the recurring decimal, $1.\dot{2}\dot{3}$ to a fraction (rational number).
(*show all working – as if not using a calculator*)

(2)

- c) Express without using negative or fractional powers:

(3)

i) x^{-3} ii) $(y + 4)^{-\frac{2}{3}}$

- d) Using your calculator, evaluate, correct to 2 decimal places:

$$\frac{72.094 \times 116.8}{79.64^2 - 16.78^2}$$

(2)

- e) The speed of light is 299 725 km/sec. Write this number correct to the nearest hundred.

(1)

- f) The distance from the sun to the earth is 149 492 000 km. Write this in scientific notation, correct to three significant figures.

(2)

Question 2**(12 marks)**

a) Simplify:

i)
$$\frac{a-6b}{2a-12b}$$

(1)

ii)
$$\frac{x^2+4x-5}{x^2-25}$$

(2)

b) Factorize:

i)
$$2x^2 + 3x - 2$$

(2)

ii)
$$8a^3 - 27$$

(2)

c) Simplify:

i) $\frac{2}{a} + \frac{5}{a+b}$

(2)

ii) $\frac{2a-3b}{5} \div \frac{4a^2-9b^2}{15}$

(3)

Question 3**(5 marks)**

a) Evaluate:

i) $4|-3| \div |-2|$

(1)

ii) $3 - |-2| + 5|4 \times -3|$

(1)

iii) $|ab|$ when $a = -5$ and $b = 6$

(1)

b) Solve:

$|x - 2| = 3$

(2)

Question 4**(9 marks)**

Simplify:

a) $\sqrt{63}$

(2)

b) $4\sqrt{96} + 3\sqrt{6} - 5\sqrt{54}$

(3)

c) If $a = \frac{\sqrt{3-1}}{\sqrt{2}}$ and $b = \frac{\sqrt{3+1}}{\sqrt{2}}$ then find: (*rationalize denominators if needed*)

i. **ab**

(2)

ii. $a - b$

(2)

Question 5**(9 marks)**

a) Solve:

i) $\frac{x-3}{3} - \frac{x+5}{2} = 4$ (3)

ii) $2a - 6 \leq 5a - 3$.
Show your answer on a number line.

(3)

b) Mark on a number line the values of x for which $|3x - 2| \leq 4$.

(3)

(46 marks)Write the answer to each question on *the question paper*. Show all working.**Question 1****(11 marks)**

- a) Find the value of
- $13^{-1.3}$
- correct to three decimal places.

$$0.036 \checkmark$$

(1)

- b) Convert the recurring decimal,
- $1.\dot{2}\dot{3}$
- to a fraction (rational number).
-
- (show all working – as if
- not
- using a calculator)

(2)

$$\text{Let } x = 1.2323\ldots$$

$$100x = 123.2323\ldots \checkmark$$

$$- x = 1.2323\ldots$$

$$\hline 99x = 122.0000$$

$$x = \frac{122}{99} \checkmark$$

- c) Express without using
- negative
- or
- fractional
- powers:

(3)

i) x^{-3}

$$= \frac{1}{x^3} \checkmark$$

ii) $(y+4)^{-\frac{2}{3}}$

$$= \frac{1}{(y+4)^{\frac{2}{3}}} \checkmark$$

$$= \frac{1}{\sqrt[3]{(y+4)^2}} \checkmark$$

d) Using your calculator, evaluate, correct to 2 decimal places:

$$\frac{(72.094 \times 116.8)}{(79.64^2 - 16.78^2)}$$

$$= 1.38931$$

$$\approx 1.39$$

(2)

e) The speed of light is 299 725 km/sec. Write this number correct to the nearest hundred.

$$299\,700 \text{ km/sec}$$

(1)

f) The distance from the sun to the earth is 149 492 000 km. Write this in scientific notation, correct to three significant figures.

$$1.49492000 \times 10^8$$

$$\therefore 1.49 \times 10^8 \quad (3 \text{ s.f.})$$

(2)

Question 2

(12 marks)

a) Simplify:

$$\begin{aligned} \text{i) } \frac{a-6b}{2a-12b} &= \frac{(a-6b)}{2(a-6b)} \\ &= \frac{1}{2} \end{aligned} \quad (1)$$

$$\begin{aligned} \text{ii) } \frac{x^2+4x-5}{x^2-25} &= \frac{(x-1)(x+5)}{(x-5)(x+5)} \\ &= \frac{x-1}{x-5} \end{aligned} \quad (2)$$

b) Factorize:

$$\begin{aligned} \text{i) } 2x^2 + 3x - 2 &= 2x^2 + 4x - x - 2 \\ &= 2x(x+2) - (x+2) \\ &= (x+2)(2x-1) \end{aligned} \quad (2)$$

$$\begin{aligned} \text{ii) } 8a^3 - 27 &= (2a)^3 - 3^3 \\ &= (2a-3)(4a^2+6a+9) \end{aligned} \quad (2)$$

c) Simplify:

$$\begin{aligned}
 \text{i)} \quad \frac{2}{a} + \frac{5}{a+b} &= \frac{2(a+b) + 5a}{a(a+b)} \checkmark \\
 &= \frac{2a + 2b + 5a}{a(a+b)} \checkmark \\
 &= \frac{7a + 2b}{a(a+b)} \checkmark
 \end{aligned}
 \tag{2}$$

$$\begin{aligned}
 \text{ii)} \quad \frac{2a-3b}{5} \div \frac{4a^2-9b^2}{15} &= \frac{2a-3b}{5} \div \frac{(2a-3b)(2a+3b)}{15} \checkmark \\
 &= \frac{(2a-3b)}{5} \times \frac{3 \cancel{15}}{(2a-3b)(2a+3b)} \checkmark \\
 &= \frac{3}{(2a+3b)} \checkmark \checkmark
 \end{aligned}
 \tag{3}$$

Question 3

(5 marks)

a) Evaluate:

i) $4|-3| \div |-2|$

$$\begin{aligned} &= 4 \times 3 \div 2 \\ &= 12 \div 2 \\ &= 6 \quad \checkmark \end{aligned}$$

(1)

ii) $3 - |-2| + 5|4 \times -3|$

$$\begin{aligned} &= 3 - 2 + 5(12) \\ &= 1 + 60 \quad \checkmark \\ &= 61 \end{aligned}$$

(1)

iii) $|ab|$ when $a = -5$ and $b = 6$

$$|-5 \times 6| = 30 \quad \checkmark$$

(1)

b) Solve:

$$|x - 2| = 3$$

(2)

$$\begin{aligned} x - 2 &= 3 & \text{or} & & -(x - 2) &= 3 \\ x &= 5 \quad \checkmark & & & -x + 2 &= 3 \\ & & & & -x &= 1 \\ & & & & x &= -1 \quad \checkmark \end{aligned}$$

Question 4

(9 marks)

Simplify:

a) $\sqrt{63}$

$$= \sqrt{9 \times 7}$$

$$= 3\sqrt{7}$$

(2)

b) $4\sqrt{96} + 3\sqrt{6} - 5\sqrt{54}$

$$= 4\sqrt{16 \times 6} + 3\sqrt{6} - 5\sqrt{6 \times 9}$$

$$= 4 \times 4\sqrt{6} + 3\sqrt{6} - 5 \times 3\sqrt{6}$$

$$= 16\sqrt{6} + 3\sqrt{6} - 15\sqrt{6}$$

$$= 4\sqrt{6} \checkmark$$

(3)

c) If $a = \frac{\sqrt{3-1}}{\sqrt{2}}$ and $b = \frac{\sqrt{3+1}}{\sqrt{2}}$ then find: (rationalize denominators if needed)

i. ab

$$\frac{\sqrt{3-1}}{\sqrt{2}} \times \frac{\sqrt{3+1}}{\sqrt{2}} \checkmark$$

(2)

$$= \frac{\sqrt{2} \times \sqrt{4}}{2} = \frac{\sqrt{2} \times 2}{2} = \sqrt{2} \checkmark$$

ii. $a - b$

$$\frac{\sqrt{3-1}}{\sqrt{2}} - \frac{\sqrt{3+1}}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} \checkmark$$

(2)

$$= \frac{\sqrt{2}}{\sqrt{2}} - \sqrt{2} \cdot \sqrt{4}$$

$$= 1 - 2\sqrt{2} \checkmark$$

Question 5

(9 marks)

a) Solve:

$$i) \quad \frac{x-3}{3} - \frac{x+5}{2} = 4$$

(3)

$$2(x-3) - 3(x+5) = 4 \times 6 \quad \checkmark$$

$$2x - 6 - 3x - 15 = 24$$

$$-x - 21 = 24 \quad \checkmark$$

$$-x = 45$$

$$x = -45 \quad \checkmark$$

$$ii) \quad 2a - 6 \leq 5a - 3.$$

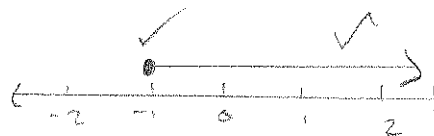
Show your answer on a number line.

$$-3 \leq 3a \quad \checkmark$$

$$-1 \leq a \quad \checkmark$$

$$\therefore a \geq -1$$

(3)

b) Mark on a number line the values of x for which $|3x - 2| \leq 4$.

$$3x - 2 \leq 4$$

$$\text{or } -(3x - 2) \leq 4$$

$$3x \leq 6$$

$$x \leq 2$$

$$\checkmark \quad -3x + 2 \leq 4$$

$$-3x \leq 2$$

$$x \geq -\frac{2}{3}$$

$$\therefore -\frac{2}{3} \leq x \leq 2 \quad \checkmark$$

(3)

